RECRUITMENT PIPELINE DATASET

Professional English Project Guide & Analysis Framework

Prepared from the **Recruitment_Pipeline** worksheet in the supplied Excel workbook. The structure follows the section sequence and practical project-guide format of the Employee Attrition Dataset document.

Dataset scope: 20,000 candidate records and 11 columns. This guide focuses only on the Recruitment_Pipeline worksheet.

1. What Does This Dataset Contain?

The Recruitment_Pipeline worksheet records candidates as they move through the hiring process, from application to offer and joining. It includes candidate background, department and role, recruitment source, current pipeline stage, key dates, and final outcome.

Column	Meaning
Candidate_ID	Unique identifier assigned to each candidate.
Department	Department recruiting for the position, such as Technology, Sales, Finance, or Customer Support.
Job_Role	Role for which the candidate has applied.
Candidate_Type	Candidate category: Fresher or Experienced.
Experience_Years	Number of years of experience recorded for the candidate.
Recruitment_Source	Channel through which the candidate entered the pipeline, such as Company Website, Campus Hiring, Employee Referral, or Professional Network.
Current_Stage	Stage currently recorded for the candidate, such as Screening, Interview, Offer, Hired, or Withdrawn.
Application_Date	Date on which the candidate applied.
Offer_Date	Date on which an offer was made, when applicable.
Joining_Date	Recorded joining date, when applicable.
Outcome	Recorded pipeline outcome, such as In Progress, Joined, or Closed.

Note: Date fields may be blank for candidates who have not reached the relevant stage. Interpret blank values in the context of the candidate's stage and outcome.

2. What Analysis Can Be Performed?

2.1 Recruitment Pipeline Analysis

• How many candidates are in the pipeline?
• How many candidates are at each current stage?
• How many candidates have joined, remain in progress, or are closed?
Useful columns: Current_Stage, Outcome, Candidate_ID, Department, Job_Role.

2.2 Recruitment Source & Candidate Analysis

• Which recruitment sources bring in the most candidates?
• Do candidate volumes differ by fresher and experienced categories?
• Which departments and job roles attract the most applications?
Useful columns: Recruitment_Source, Candidate_Type, Experience_Years, Department, Job_Role.

2.3 Hiring Timeline & Workforce Readiness

• How long does it take to move from application to offer?
• How long is the recorded period between application and joining?
• Which roles have candidates waiting at later stages?
Useful columns: Application_Date, Offer_Date, Joining_Date, Current_Stage, Department, Job_Role.

2.4 Recruitment Outcome Analysis

• What proportion of recorded candidates have joined?
• How do recorded outcomes vary by department, job role, candidate type, or recruitment source?
• At which stages are candidates most frequently recorded as withdrawn or closed, where those values are available?
Useful columns: Outcome, Current_Stage, Recruitment_Source, Candidate_Type, Department, Job_Role.

3. Which Columns Are Most Important for the Project?

A. Recruitment Funnel Reporting
Current_Stage, Outcome, Department, Job_Role, Recruitment_Source.

B. Hiring Timeline Analysis
Application_Date, Offer_Date, Joining_Date, Current_Stage, Outcome.

C. Candidate Segmentation
Candidate_Type, Experience_Years, Department, Job_Role, Recruitment_Source.

D. Outcome Prediction — Optional Machine Learning Task
Potential input features may include Candidate_Type, Experience_Years, Department, Job_Role, and Recruitment_Source. The target could be a clearly defined outcome such as Joined versus Not Joined, but only after confirming the meaning and completeness of the recorded Outcome values.

Important points
• Candidate_ID is an identifier and should normally not be used as a predictive input feature.
• Current_Stage, Offer_Date, and Joining_Date may reveal information that would not have been available at the time a prediction is intended to be made. Exclude later-stage information when building an early-stage prediction model to avoid target leakage.
• The workbook README states that the records are synthetic and created for practice. Findings should not be presented as evidence of actual company recruitment performance.

4. What New Calculated Columns Can Be Created?

New column	Suggested calculation	Purpose
Days_To_Offer	Offer_Date – Application_Date	Measure the elapsed days from application to offer; calculate only when both dates are present.
Days_To_Join	Joining_Date – Application_Date	Measure the elapsed days from application to joining; calculate only when both dates are present.
Days_Offer_To_Join	Joining_Date – Offer_Date	Measure the interval between offer and joining; calculate only when both dates are present.
Application_Month	Month extracted from Application_Date	Monthly application trend analysis.
Application_Year	Year extracted from Application_Date	Year-wise recruitment trend analysis.
Experience_Group	Group Experience_Years into defined bands	Compare pipeline outcomes across experience ranges.
Candidate_Type_Flag	Encode Fresher / Experienced as categories or numeric indicators	Prepare candidate type for analysis or model preprocessing.
Joined_Flag	Joined = 1; other defined outcomes = 0	Use only after defining the exact comparison group and handling In Progress records separately.
Stage_Order	Map each approved pipeline stage to a documented order	Visualize candidate progression; confirm the organization's actual stage sequence first.

Date-based calculations depend on valid date values. Keep missing dates blank where the event has not occurred; do not automatically treat them as zero-day durations.

5. What Advanced Metrics Can Be Created?

Metric	Calculation / definition	Use
Candidate Count	Count of distinct Candidate_ID values.	Overall pipeline volume.
Stage Distribution	Candidates at each Current_Stage ÷ total candidates × 100.	Understand the pipeline mix.
Recorded Joining Rate	Candidates with Outcome = Joined ÷ a clearly defined eligible candidate group × 100.	Summarize recorded joining outcomes; define the denominator explicitly.
Source-wise Joining Rate	Joined candidates from a source ÷ eligible candidates from that source × 100.	Compare recruitment sources fairly.
Average Days to Offer	Average Days_To_Offer for candidates with valid application and offer dates.	Review offer turnaround time.
Average Days to Join	Average Days_To_Join for candidates with valid application and joining dates.	Review end-to-end recorded hiring time.
Open Pipeline Count	Count candidates whose Outcome is In Progress, after validating category definitions.	Monitor candidates still being processed.

Metrics should use a documented denominator and consistent date rules. A high candidate count or joining rate alone does not establish recruitment quality or business impact.

6. Simple Project Workflow

1. Review the Recruitment_Pipeline columns and confirm the meaning of each stage and outcome.

2. Check missing values, duplicate Candidate_ID values, date formats, invalid date sequences, and inconsistent category labels.

3. Keep blank Offer_Date and Joining_Date values where those events have not occurred; investigate unexpected blanks separately.

4. Create suitable calculated columns such as Days_To_Offer, Days_To_Join, Application_Month, and Experience_Group.

5. Use Excel or Power BI to build a recruitment dashboard with candidate count, stage distribution, source-wise applications, outcomes, and hiring timelines.

6. Use SQL to aggregate candidate volumes and outcomes by department, role, candidate type, recruitment source, and time period.

7. If appropriate, build and evaluate an ML model using a clearly defined target and only information available at the intended prediction point.

8. Present the results with clear definitions, limitations, and practical recommendations.

7. Machine Learning — How Could It Be Used in This Project?

7.1 Main Purpose

A machine-learning model could learn patterns from historical candidate records and estimate a defined recruitment outcome for new records. For example, the target could be whether a candidate is recorded as Joined, provided the outcome labels are complete and suitable for that task.

7.2 A Practical Approach

• Define the prediction question and the eligible candidate population.
• Inspect the values and meaning of Outcome before converting it into a target label.
• Select only features available at the point when the prediction would be made.
• Split the data into training and test sets and evaluate performance with appropriate classification metrics.
• Compare model results with a simple baseline and explain limitations.

7.3 Interpretation and Limitations

A model estimate is not a guarantee of a candidate's future decision. Since the supplied workbook uses synthetic practice records, model performance on this dataset does not prove real-world predictive performance. Treat results as a learning exercise and validate the approach on suitable real-world data before operational use.

Final scope note: This guide uses only the Recruitment_Pipeline worksheet's fields and the workbook's stated synthetic-data limitation. It does not combine data from the Employees, Notice_Period, Recruitment_Costs, or Workforce_Planning worksheets.